Coverage Correlation Gaps under Graph-Constrained Pairwise Independence

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How much can linearly many pairwise moment constraints reduce the best expected coverage attainable with prescribed marginals? For every integer $k \geq 1$, we determine the exact ratio supremum over both k-colorable graphs and graphs of maximum degree at most $k-1$: it is $\left[1-(1-1/k)^k\right]^{-1}$. The same suprema hold when each tested joint probability may exceed the product of its marginals by any fixed multiplicative factor.

Tightness uses disjoint k-cliques with $n(k-1)/2$ tested pairs. A pointwise collision certificate controls every feasible coupling while leaving all cross-clique pairs unrestricted. It also yields heterogeneous and additive error bounds. For the matching witness we give exact finite optima and necessary and sufficient error scales for its limiting gap; five small examples have rational primal and dual certificates. Finally, we derive a finite full t-wise independence bound and an explicit rate to the previously known $e/(e-1)$ limit. The argument combines known color-resampling and Cartesian-coverage ingredients with a certificate that survives deletion of the cross-block moment constraints.

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1 Sparse moment information

Can $O(n)$ prescribed pairwise moments enforce a constant loss in the best expected coverage, even when almost all pairs remain unrestricted? Consider a coupling that selects an entire block: it can cover every feature while keeping each vertex's marginal small. Testing pairs within blocks restricts precisely the collisions that make this coupling valuable. We quantify the resulting loss, identifying a sharp hierarchy indexed by a coloring bound. The information consists of exact probability constraints; statistical estimation of those probabilities is a separate question.

Model. For a finite ground set V, a simple graph $G = (V, E)$, and $x \in [0,1]^V$, let $\mathcal{D}(x)$ be all laws of $S \subseteq V$ with $\mathbb{P}(i \in S) = x_i$. Define

$$\mathcal{D}_G^{\mathrm{eq}}(x) = \{D \in \mathcal{D}(x) : \mathbb{P}_D(i, j \in S) = x_i x_j \ (ij \in E)\},$$
$$\mathcal{D}_{G,\delta}^{\mathrm{up}}(x) = \{D \in \mathcal{D}(x) : \mathbb{P}_D(i, j \in S) \le (1+\delta) x_i x_j \ (ij \in E)\},$$

where $\delta \ge 0$ is finite. The graph specifies tested pairs: nonedges impose no restrictions, unlike a dependency-graph convention. In particular $\mathcal{D}_G^{\mathrm{eq}}(x) \subseteq \mathcal{D}_{G,0}^{\mathrm{up}}(x)$, since upper constraints also permit negative correlations. A nonnegative weighted coverage function has the form

$$f(S) = \sum_{H \in \mathcal{H}} w_H \mathbf{1}\{S \cap H \neq \varnothing\}, \qquad w_H \ge 0,$$

with a finite feature family and finite weights. Set

$$f^+(x) = \max_{D \in \mathcal{D}(x)} \mathbb{E}_D f(S), \quad f_G^{\mathrm{eq}}(x) = \max_{D \in \mathcal{D}_G^{\mathrm{eq}}(x)} \mathbb{E}_D f(S), \quad f_{G,\delta}^{\mathrm{up}}(x) = \max_{D \in \mathcal{D}_{G,\delta}^{\mathrm{up}}(x)} \mathbb{E}_D f(S).$$

All maxima exist: their feasible sets are nonempty closed subsets of a finite probability simplex, containing the product law. We consider ratios with $f^+(x) > 0$; a positive-weight feature then contains a positive-marginal vertex, so the product expectation and every denominator are positive.

For a class $\mathcal{A}$ of finite graphs, write

$$\Gamma^{\mathrm{eq}}(\mathcal{A}) = \sup_{G \in \mathcal{A},\, x, f:\, f^+(x) > 0} \frac{f^+(x)}{f_G^{\mathrm{eq}}(x)}, \qquad \Gamma_\delta^{\mathrm{up}}(\mathcal{A}) = \sup_{G \in \mathcal{A},\, x, f:\, f^+(x) > 0} \frac{f^+(x)}{f_{G,\delta}^{\mathrm{up}}(x)}.$$

The ground set, marginals and finite feature family vary in these suprema. Let $\mathcal{C}_k = \{G : \chi(G) \le k\}$, $\mathcal{B}_k = \{G : \Delta(G) \le k-1\}$, and $\alpha_k = 1 - (1 - 1/k)^k$.

Theorem 1.1. *For every integer $k \ge 1$ and every fixed finite $\delta \ge 0$,*

$$\Gamma^{\mathrm{eq}}(\mathcal{C}_k) = \Gamma^{\mathrm{eq}}(\mathcal{B}_k) = \Gamma_\delta^{\mathrm{up}}(\mathcal{C}_k) = \Gamma_\delta^{\mathrm{up}}(\mathcal{B}_k) = \alpha_k^{-1}.$$

These are class suprema, approached by a sequence of finite witnesses. For $k = 2$, a matching on $2m$ vertices suffices: its specified witness has exact equality-model ratio $4m/(3m+1)$, tending to $4/3$ (Theorem 4.1). For fixed k, the witnesses use $n(k-1)/2$ tested pairs. Table 1 records the first constants and the finite-error coefficient.

Table 1. Exact constants. The witness's multiplicative finite upper bound is $\min\{1, \alpha_k + c_k(1+\delta)/m\}$, with $c_k = (k-1)/(2k)$.

k	2	3	4	5	6
α_k	3/4	19/27	175/256	2101/3125	31031/46656
α_k^{-1}	4/3	27/19	256/175	3125/2101	46656/31031
c_k	1/4	1/3	3/8	2/5	5/12

Two complementary ingredients. A coloring converts any marginal-preserving sampler into a feasible law, retaining an α_k fraction of its coverage. This gives the class upper bound. For sharpness we use a known Cartesian coverage family on disjoint cliques. The obstacle is to control *every* coupling after cross-block tests have been removed: deleting tests enlarges the denominator. Our pointwise certificate uses only total occupancy and within-block collisions, whose expectations are fixed or bounded by the sparse tests. It also yields error tolerance, exact matching examples and a finite independence-hierarchy bound.

2 Relation to prior work

Agrawal, Ding, Saberi and Ye [ADSY12] developed correlation-gap bounds for stochastic optimization, comparing the unrestricted coupling with the mutually independent expectation. The multilinear extension and continuous-greedy framework of Calinescu, Chekuri, Pál and Vondrák [CCPV11] provide a related foundation in submodular optimization under matroid constraints. Our denominator instead maximizes over laws subject to a prescribed collection of moment constraints.

Ramachandra and Natarajan [RN25] introduced the pairwise-independent correlation-gap framework. Their three-variable pairwise-optimized bound is $4/3$ (Theorem 5); the dimension-dependent $27/19$ value discussed in their Section 1.1 uses the *mutually* independent denominator. These constants concern different feasible sets. Their later five-element coverage counterexample [RN26] gives pairwise ratio at least $640/479 > 4/3$, disproving the proposed universal pairwise $4/3$ bound while preserving the three-variable result. Here k measures graph colorability, and the number of variables is $n = mk$.

Ramachandra [Ram26, Theorem 2 and Section 4] gives the Cartesian coverage family used below, its unrestricted optimum and the asymptotic all-pairs gap $e/(e-1)$, also noting the t-wise consequence. Our certificate keeps the block size k fixed and allows arbitrary cross-block dependence. It proves the sparse tightness needed for Theorem 1.1; Corollary 5.1 also recovers the known limit with an explicit finite bound.

The two-group coverage calculation in Section 3.2 of the full version of *The Cost of Consistency*, by Paul Dütting, Federico Fusco, Silvio Lattanzi, Ashkan Norouzi-Fard, Ola Svensson and Morteza Zadimoghaddam [DFL+25], gives the $3/4$ resampling factor. Lemma 3.1 is its elementary k-group version. Udwani and Wajc [UW26, Section 3.1] use the Cartesian coverage polynomial, α_k, and a no-collision specialization in online submodular welfare. Our collision penalty quantifies the effect of allowing collisions subject to tested moments. Thus the contribution combines sparse tightness with a sharp graph-class hierarchy and finite error refinements; the construction, constant, resampling ingredient and asymptotic full-independence limit have the preceding origins.

3 Resampling and a collision certificate

3.1 A conversion from a coloring

Lemma 3.1. *Given a proper k-coloring of G and sampler access to a law $D \in \mathcal{D}(x)$, k independent calls produce a law in $\mathcal{D}_G^{\mathrm{eq}}(x)$ with expected coverage at least $\alpha_k \mathbb{E}_D f$, simultaneously for all nonnegative weighted coverage functions. Consequently $f_G^{\mathrm{eq}}(x) \geq \alpha_k f^+(x)$.*

Proof. For $k \geq 2$, partition $V = C_1 \sqcup \cdots \sqcup C_k$ into independent color classes, allowing empty classes. Draw independent $S_1, \ldots, S_k$ with law D, and return $R = \bigcup_c (S_c \cap C_c)$. Each vertex keeps its marginal; edge endpoints use different independent calls, so their joint probability is $x_i x_j$.

For a feature H, put $a_c = \mathbb{P}_D(S \cap H \cap C_c \neq \varnothing)$, $s = \sum_c a_c$, and $b = \mathbb{P}_D(S \cap H \neq \varnothing)$. The union bound gives $b \leq \min\{1, s\}$. By independence and AM–GM,

$$\mathbb{P}(R \cap H \neq \varnothing) = 1 - \prod_c (1 - a_c) \geq h(s) := 1 - (1 - s/k)^k.$$

On $[0, k]$, h is increasing and concave, with $h(0) = 0$ and $h(1) = \alpha_k$. Its chord on $[0, 1]$ and monotonicity thereafter give $h(s) \geq \alpha_k \min\{1, s\} \geq \alpha_k b$. Multiplying by nonnegative weights and summing proves the claim for the single common output law. Apply it to a maximizing input law for the last assertion. When $k = 1$, the graph is edgeless and the unchanged input law suffices. □

The conversion uses $O(k|V|)$ membership operations and no feature queries. Its input is a sampler; obtaining a maximizing law for f^+ is a distinct optimization problem.

3.2 The Cartesian witness

Fix $k \geq 2$ and $m \geq 1$. Take disjoint blocks $B_1, \ldots, B_m$ of k vertices, and test all pairs within each block. For every $u \in \{1, \ldots, k\}^m$, give the transversal feature $H_u = \{(r, u_r) : 1 \leq r \leq m\}$ weight k^{-m}. For $a_r = |S \cap B_r|$, counting missed features gives

$$f(S) = 1 - \prod_{r=1}^{m} (1 - a_r/k). \tag{1}$$

Here *normalized* means $f(\varnothing) = 0$, $f(V) = 1$ and $0 \leq f(S) \leq 1$. The explicit feature family has k^m members; (1) evaluates it from block counts in $O(n + m)$ arithmetic operations. Set $x_i = 1/m$. Selecting one whole block uniformly has these marginals and hits every transversal, so $f^+(x) = 1$. The tested graph has $n = mk$ vertices and $|E| = m\binom{k}{2} = n(k-1)/2$, with $\chi(G) = k$ and $\Delta(G) = k - 1$. Figure 1 shows the construction and the statistic that the moment constraints control.

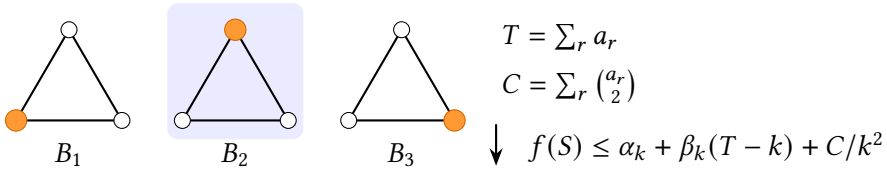


Fig. 1. The witness for $k = m = 3$. Solid triangle edges are tested pairs; there are no cross-block tests. Orange marks identify one transversal feature, a set of vertices rather than graph edges. Shading selects the whole block B_2, which hits every transversal. Such draws have $C = 3$, while equality-tested moments force $\mathbb{E}C = 1$. The certificate uses only $T = |S|$ and C.

3.3 A pointwise bound and all error models

Lemma 3.2. *Let $q = 1 - 1/k$, $\beta_k = q^k/k$, $T = |S|$, and $C = \sum_r \binom{|S \cap B_r|}{2}$. Every subset satisfies*

$$f(S) \leq \alpha_k + \beta_k(T - k) + C/k^2. \tag{2}$$

Proof. For integers $0 \leq a \leq k$, let $d_a = q^a - (1 - a/k)$. Then $d_0 = 0$ and

$$d_{a+1} - d_a = (1 - q^a)/k, \qquad 0 \leq 1 - q^a = k^{-1} \sum_{j=0}^{a-1} q^j \leq a/k.$$

Summing gives $0 \leq d_a \leq \binom{a}{2}/k^2$. Set $A_r = 1 - a_r/k$ and $Q_r = q^{a_r}$, so $0 \leq A_r \leq Q_r \leq 1$. The product identity

$$\prod_r Q_r - \prod_r A_r = \sum_r (Q_r - A_r) \prod_{s<r} Q_s \prod_{s>r} A_s \leq C/k^2$$

yields $f(S) \leq 1 - q^T + C/k^2$. For integer $t \geq 0$, $g(t) = 1 - q^t$ has decreasing successive differences q^t/k. Summing differences above k, or backwards below k, gives $g(t) \leq g(k) + \beta_k(t - k)$. Combining the inequalities proves (2). □

For edge budgets $e_{ij} \geq 0$, define $f^{\mathrm{up}}_{G,e}(x)$ by replacing edge equalities with $\mathbb{P}(i, j \in S) \leq x_i x_j + e_{ij}$. Write $f^{\mathrm{add}}_{G,\epsilon}$ for the uniform choice $e_{ij} = \epsilon$.

Theorem 3.3. *For the witness in Section 3.2 and any finite nonnegative edge budgets,*

$$\alpha_k \leq f^{\mathrm{up}}_{G,e}(x) \leq \min\left\{1, \alpha_k + \frac{k-1}{2km} + \frac{1}{k^2} \sum_{ij \in E} e_{ij}\right\}. \tag{3}$$

In particular, for finite $\epsilon, \delta \geq 0$,

$$\alpha_k \leq f^{\mathrm{add}}_{G,\epsilon}(x) \leq \min\{1, \alpha_k + \tfrac{k-1}{2k}(1/m + m\epsilon)\}, \tag{4}$$

$$\alpha_k \leq f^{\mathrm{up}}_{G,\delta}(x) \leq \min\{1, \alpha_k + \tfrac{(1+\delta)(k-1)}{2km}\}. \tag{5}$$

Also $\alpha_k \leq f^{\mathrm{eq}}_G(x) \leq \min\{1, \alpha_k + (k-1)/(2km)\}$.

Proof. Every admissible law has $\mathbb{E}T = mk/m = k$, and

$$\mathbb{E}C = \sum_{ij \in E} \mathbb{P}(i, j \in S) \leq \frac{m\binom{k}{2}}{m^2} + \sum_{ij \in E} e_{ij}.$$

Average (2); its linear term cancels and its collision coefficient is nonnegative. Together with $f \leq 1$, this proves the upper bound. Lemma 3.1 and inclusion of equality-feasible laws give the lower bound. Substituting $e_{ij} = \epsilon$ or δ/m^2 gives (4) and (5); equality laws are feasible for the zero-error upper model, giving the final assertion. At $m = 1$, all vertices are forced present, and every displayed denominator equals one. □

3.4 Completing the class theorem

Proof of Theorem 1.1. For $k = 1$ every graph in either class is edgeless, giving ratio one. For $k \geq 2$, greedy coloring gives $\mathcal{B}_k \subseteq \mathcal{C}_k$. Feasible-set inclusion and Lemma 3.1 give the two chains

$$\Gamma^{\mathrm{up}}_\delta(\mathcal{B}_k) \leq \Gamma^{\mathrm{eq}}(\mathcal{B}_k) \leq \Gamma^{\mathrm{eq}}(\mathcal{C}_k) \leq 1/\alpha_k,$$

$$\Gamma^{\mathrm{up}}_\delta(\mathcal{B}_k) \leq \Gamma^{\mathrm{up}}_\delta(\mathcal{C}_k) \leq \Gamma^{\mathrm{eq}}(\mathcal{C}_k) \leq 1/\alpha_k.$$

The witness belongs to $\mathcal{B}_k$ and, by (5), has upper-model ratio at least $[\alpha_k + (1+\delta)(k-1)/(2km)]^{-1}$. Letting $m \to \infty$ at fixed k, δ makes the endpoints of both chains equal, proving all four identities. □

4 Exact finite examples and error scales

4.1 A matching witness with an exact transition

Theorem 4.1. *For the Cartesian witness with $k = 2$, every integer $m \geq 1$ and finite $\delta, \epsilon \geq 0$,*

$$f^{\mathrm{eq}}_G(x) = \frac{3}{4} + \frac{1}{4m}, \qquad \frac{f^+(x)}{f^{\mathrm{eq}}_G(x)} = \frac{4m}{3m+1}, \tag{6}$$

$$f^{\mathrm{up}}_{G,\delta}(x) = \frac{3}{4} + \frac{1}{4}\min\{1, (1+\delta)/m\}, \tag{7}$$

$$f^{\mathrm{add}}_{G,\epsilon}(x) = \frac{3}{4} + \frac{1}{4}\min\{1, 1/m + m\epsilon\}. \tag{8}$$

For any nonnegative error sequences as $m \to \infty$, these latter denominators tend to $3/4$ if and only if $\delta_m/m \to 0$, respectively $m\epsilon_m \to 0$. If the respective scaled error tends to $c \in [0, \infty]$, the denominator tends to $3/4 + \min\{1, c\}/4$ and the ratio to $4/(3 + \min\{1, c\})$.

Proof. For $m \geq 2$, type A selects two distinct uniform blocks and one independent fair vertex in each: its marginals are $1/m$, edge moments zero, and coverage $3/4$. Type B selects one entire uniform block: it has the same marginals, edge moments $1/m$, and coverage one. Using type B with probability θ gives moments θ/m and value $3/4 + \theta/4$. For the multiplicative model choose $\theta = \min\{1, (1+\delta)/m\}$; for the additive model choose $\theta = \min\{1, 1/m + m\epsilon\}$. Their values meet the respective bounds of Theorem 3.3. For equality take $\theta = 1/m$, meeting the zero-error upper bound with each edge equality exact. For $m = 1$, the only feasible set is V, giving one in every formula. The limit statements follow directly from the formulas and continuity of $\min\{1, u\}$ on the extended nonnegative half-line. Necessity follows because $\min\{1, u_m\} \to 0$ forces $u_m \to 0$ for $u_m \geq 0$. □

General sufficient scales. For every fixed $k \geq 2$, Theorem 3.3 implies convergence of the witness denominator to α_k and of its ratio to $1/\alpha_k$ whenever $\delta_m = o(m)$, or, in the additive model, $\epsilon_m = o(1/m)$. More

generally $\sum_E e_{ij}^{(m)} \to 0$ suffices. The per-edge absolute excess in the multiplicative model is δ_m/m^2; there are $m\binom{k}{2}$ tested edges. This explains the aggregate scale. The necessity assertions of Theorem 4.1 apply to its specified matching witness and uniform marginals.

4.2 Five rational optimality certificates

A profile is the sorted vector of block occupancies. Averaging a law over block permutations and independent within-block vertex permutations preserves coverage and all uniform constraints. The group is transitive on vertices and on tested edges, so a uniform profile-orbit mixture has marginal $\mathbb{E}T/(mk)$ at each vertex and edge moment $\mathbb{E}C/[m\binom{k}{2}]$. Thus $\mathbb{E}T = k$ and the relevant constraint on $\mathbb{E}C$ suffice for such a mixture to lift to a feasible law.

Proposition 4.2. *The four equality optima at* $(k, m) = (2, 2), (2, 3), (3, 2), (3, 3)$ *are* $7/8, 5/6, 5/6, 65/81$, *respectively. At* $(3, 3)$ *the upper-model optimum with* $\delta = 1$ *is* $73/81$.

Proof. Table 2 supplies orbit mixtures and pointwise upper bounds $f(S) \le y_0 + y_T T + y_C C$. Their masses sum to one, $\mathbb{E}T = k$, and $\mathbb{E}C = k(k-1)/(2m)$ for equality, or twice that quantity in the last row. Their objectives are the listed values. For $k = 2$ the pointwise inequality holds as follows: if $C = 0$, then $f = 1 - 2^{-T} \le 1/4 + T/4$, checked at $T = 0, 1, 2$ and immediate for $T \ge 3$; if $C \ge 1$, a full block gives $f = 1$, $T \ge 2$, and the asserted upper bound is at least one. For $k = 3, m = 2$, the exhaustive profiles in order

$$(0, 0), (0, 1), (0, 2), (0, 3), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)$$

have dual slacks $1/3, 1/9, 0, 0, 0, 0, 1/9, 1/9, 1/3, 2/3$. For $k = m = 3$, Table 3 exhausts all 20 profiles; (1) depends only on the profile, so these cover all 512 labeled subsets. Every slack is nonnegative. Averaging the dual bound matches the primal values in each row; $y_C \ge 0$ justifies its use with the upper constraint. Finite weak duality proves exact optimality. □

Table 2. Exact primal/dual certificates. Each profile denotes its uniform subset orbit. All omitted masses are zero; the last row uses $\delta = 1$.

(k, m)	Model	Primal profiles: masses	(y_0, y_T, y_C)	Optimum
$(2, 2)$	eq	$(0, 2) : 1/2;\ (1, 1) : 1/2$	$(1/4, 1/4, 1/4)$	$7/8$
$(2, 3)$	eq	$(0, 0, 2) : 1/3;\ (0, 1, 1) : 2/3$	$(1/4, 1/4, 1/4)$	$5/6$
$(3, 2)$	eq	$(0, 3) : 1/4;\ (1, 2) : 3/4$	$(1/3, 1/9, 1/9)$	$5/6$
$(3, 3)$	eq	$(0, 0, 3) : 1/3;\ (1, 1, 1) : 2/3$	$(8/27, 11/81, 8/81)$	$65/81$
$(3, 3)$	up	$(0, 0, 3) : 2/3;\ (1, 1, 1) : 1/3$	$(8/27, 11/81, 8/81)$	$73/81$

Reproducibility. The accompanying `finite-certificates.json` stores all five rational inputs and certificates. The standalone `verify_revision.py` reconstructs uniform orbit laws, checks every marginal and tested pair, and verifies the objective and dual inequality on every labeled subset using exact fractions; `discover_finite.py` records the separate rational profile search. These files accompany the source package; `manuscript/reproduce.py` provides a read-only launcher.

5 Consequences and open questions

5.1 A finite full-independence hierarchy

For the same witness, let $f_t(x)$ maximize expected coverage over laws whose indicators are jointly independent on every subfamily of at most t vertices, with $2 \le t \le n = mk$. Thus $t = 2$ tests all pairs, and $t = n$ specifies the product law.

Table 3. Complete pointwise certificate for $k = m = 3$: slack of $8/27 + (11/81)T + (8/81)C - f(S)$.

Profile	Slack	Profile	Slack
$(0,0,0)$	$8/27$	$(1,1,1)$	0
$(0,0,1)$	$8/81$	$(1,1,2)$	$7/81$
$(0,0,2)$	0	$(1,1,3)$	$22/81$
$(0,0,3)$	0	$(1,2,2)$	$20/81$
$(0,1,1)$	$1/81$	$(1,2,3)$	$41/81$
$(0,1,2)$	$2/81$	$(1,3,3)$	$68/81$
$(0,1,3)$	$11/81$	$(2,2,2)$	$4/9$
$(0,2,2)$	$4/27$	$(2,2,3)$	$20/27$
$(0,2,3)$	$10/27$	$(2,3,3)$	$29/27$
$(0,3,3)$	$19/27$	$(3,3,3)$	$38/27$

Corollary 5.1. *Let $k, m \geq 2$, $a = 1 - 1/e$, and $U = \alpha_k + (k-1)/(2km)$. Uniformly for $2 \leq t \leq mk$,*

$$\alpha_m \leq f_t(x) \leq \min\{1, U\}, \qquad \frac{1}{U} \leq \frac{f^+(x)}{f_t(x)} \leq \frac{1}{\alpha_m}, \tag{9}$$

$$0 \leq \frac{e}{e-1} - \frac{f^+(x)}{f_t(x)} \leq \frac{1/k + 1/(2m)}{(1-1/e)^2}. \tag{10}$$

Consequently any sequences $k, m \to \infty$, with any admissible choices of t, give ratio tending to $e/(e-1)$. On $k = m = s \geq 2$, the error is at most $3/[2(1-1/e)^2\sqrt{n}]$, where $n = s^2$.

Proof. The product law is feasible and hits each m-vertex feature with probability $1 - (1-1/m)^m = \alpha_m$. Every feasible t-wise law satisfies the within-clique equalities, so Theorem 3.3 bounds its objective above by $\min\{1, U\}$. Since $f^+ = 1$, this proves (9). For $j \geq 2$, $\log(1-1/j) \leq -1/j$ and $\log(1+1/(j-1)) \leq 1/(j-1)$ imply

$$a \leq \alpha_j \leq a + 1/(ej) \leq a + 1/j.$$

Thus the ratio is at most $1/a$, while $U \geq a$ and $U - a \leq 1/k + 1/(2m)$. Subtracting its lower bound $1/U$ from $1/a$ gives $(U-a)/(aU) \leq (1/k + 1/(2m))/a^2$. This proves the rate and its square-size specialization. □

The limiting theorem and family are due to Ramachandra [Ram26]; the finite estimate follows here from within-block moments alone. The product expectation is α_m, depending on the number of vertices per feature: for example $k = 2, m = 3, t = 6$ gives $19/27$.

5.2 Graph structure and optimization questions

The exact matching formulas isolate how a limited moment budget constrains a useful unrestricted coupling, and Lemma 3.1 converts an available sampler into a feasible one with a quantified loss. Together these give finite optimization consequences of the class theorem. For general fixed graphs, determining the best constant from structure beyond an integral coloring remains an open direction here. Fractional-color information and high-chromatic triangle-free graphs are natural tests. The construction gives a sufficient linear number of tested pairs at fixed k; a necessary pair budget requires a separate lower bound. Extending the sharp hierarchy to arbitrary monotone submodular objectives also remains unresolved.

Computational and AI methodology. OpenAI Codex was used for topic selection, proof development, exact computations, literature checks, drafting and typesetting. A separate model-context pass checked the proofs and finite certificates.

References

[ADSY12] Shipra Agrawal, Yichuan Ding, Amin Saberi, and Yinyu Ye. Price of correlations in stochastic optimization. *Operations Research*, 60(1):150–162, 2012. https://doi.org/10.1287/opre.1110.1011.

[CCPV11] Gruia Calinescu, Chandra Chekuri, Martin Pál, and Jan Vondrák. Maximizing a monotone submodular function subject to a matroid constraint. *SIAM Journal on Computing*, 40(6):1740–1766, 2011. https://doi.org/10.1137/080733991.

[DFL+25] Paul Dütting, Federico Fusco, Silvio Lattanzi, Ashkan Norouzi-Fard, Ola Svensson, and Morteza Zadimoghaddam. The cost of consistency: Submodular maximization with constant recourse. In *Proceedings of the 57th Annual ACM Symposium on Theory of Computing*, STOC '25, pages 1406–1417. Association for Computing Machinery, 2025. https://doi.org/10.1145/3717823.3718131. Full version: https://arxiv.org/abs/2412.02492v1.

[Ram26] Arjun Ramachandra. Dimension dependent correlation gap bounds under restricted independence. arXiv:2609.02659v1, 2026.

[RN25] Arjun Ramachandra and Karthik Natarajan. Pairwise independent correlation gap. *Operations Research Letters*, 60:107255, 2025. https://doi.org/10.1016/j.orl.2025.107255.

[RN26] Arjun Ramachandra and Karthik Natarajan. Counterexample to a conjecture on the pairwise independent correlation gap using AI. arXiv:2606.19663v1, 2026.

[UW26] Rajan Udwani and David Wajc. Online submodular welfare maximization: Computation is not the bottleneck. arXiv:2610.16060v1, 2026.